Marcel Harrer, M.Sc.

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Professional Summary

AI Engineer with 3+ years of experience building production-grade machine learning systems across computer vision, NLP, time series forecasting, and agent-based architectures. Strong background in applied mathematics (M.Sc.) with hands-on experience in Retrieval-Augmented Generation (RAG), intelligent agents, and scalable data pipelines. Proven ability to translate research into reliable, real-world AI solutions.

Education

University of Vienna
M.Sc. in Applied Mathematics — Oct 2020 – Nov 2023

University of Vienna
B.Sc. in Mathematics — Oct 2016 – Sep 2020

HTL Mödling
Technical High School — Oct 2011 – Jun 2016

Experience

Fivesquare — AI Engineer — **Remote | 2026 – Present**

- Building production AI products and infrastructure with the Fivesquare team.
- Working across LLM-based systems and agent architectures in Python.

Independent AI Engineer — Freelance — **Remote | Mar 2025 – 2026**

- Designed and deployed intelligent agent systems using LangGraph, including a Text-to-SQL agent enabling natural language access to PostgreSQL databases.
- Built modular tool interfaces supporting structured reasoning, database querying, and automated email workflows.
- Optimized agent orchestration, prompt strategies, and multi-step reasoning pipelines for reliability and accuracy.

ImageTwin — AI Research Engineer — **Jan 2024 – 2026**

- Developed computer vision algorithms to detect image manipulation and integrity violations in scientific figures.
- Implemented keypoint-based image matching using OpenCV (SIFT, SURF) and trained LightGlue models for robust visual correspondence.
- Co-developed a deep learning pipeline for AI-generated image detection, covering data annotation, preprocessing, and automated evaluation.
- Contributed to dataset creation and experimental validation in a research-driven production environment.

Financial Market Authority (FMA) — AI Engineer — **Dec 2022 – Dec 2023**

- Designed and implemented a Retrieval-Augmented Generation (RAG) system for answering domain-specific regulatory queries using FAISS and transformer-based embeddings.
- Enabled semantic search over large German-language document collections, improving information retrieval accuracy and efficiency.
- Applied SBERT-based similarity models for document clustering and regulatory text analysis.
- Produced analytical dashboards and visualizations in R to support data-driven insights in the Austrian insurance sector.

Projects

Master's Thesis – Accelerating Gradient Descent via L-Smoothness
Developed an adaptive optimization algorithm leveraging L-smoothness to dynamically adjust step sizes in gradient descent. Demonstrated provable convergence guarantees and improved convergence speed compared to fixed-step methods.

Agent Lawyer
Built a legal assistant system using Retrieval-Augmented Generation (RAG) to provide grounded, document-based answers to legal queries. Implemented semantic search with FAISS, transformer embeddings, and a reranking model to improve answer relevance. Runs locally on a compact LLaMA 3.2 model.

GitHub: github.com/MarcelHarrer/agent-lawyer-public

Technical Skills

Languages: Python, SQL, R, MATLAB

ML / AI: PyTorch, TensorFlow, Hugging Face Transformers, Computer Vision, Deep Learning, Time Series Forecasting, Retrieval-Augmented Generation (RAG), Intelligent Agents

Backend & MLOps: FastAPI, Docker, CI/CD, FAISS

Cloud & AWS: AWS SageMaker, AWS S3, AWS Lambda, AWS EC2, AWS Bedrock, AWS CloudWatch, AWS IAM

Data & Analytics: Pandas, Power BI, SQL Databases

Tools: Git, LangChain, LangGraph, OpenCV

Mathematics: Optimization, Applied Mathematics, Data Science

Languages (Spoken): German (Native), English (Bilingual), Spanish (Advanced)